CLASS 8 · JEE ADVANCED · INSTITUTE TOP 10 VS OUR TOP 10

Pragya G8 (EXI) Achievers

Part Test-1 · Paper 2 · where our toppers fall behind the institute's toppers

Exam date 09 Oct 2026
Campus HYD/Nallakunta DD Colony
Compared institute top 10 vs our top 10

On average the institute's top 10 scored ≈ 137.4 and our top 10 scored ≈ 99.7 – a gap of ≈ 37.7 marks per student: 8.5 in Maths, 15.9 in Physics and 13.3 in Chemistry.

Maths 23% of the gap | Physics 42% of the gap | Chemistry 35% of the gap

Scores are rebuilt from each question's right / wrong / skipped counts with the paper's marking; approximate (≈) because multiple-correct questions give partial marks – our official average is 95.5.

≈ 137 / 163
institute top 10 – average score

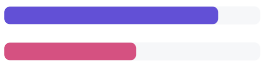
≈ 100 / 163
our top 10 – average score

11 questions
where 4+ more of the institute top 10 got it right

1 questions
where 2+ more of ours got it right

1 Subject by subject

Average marks per student, and how the answers split into right, wrong and skipped. ■ Institute top 10 ■ Our top 10

SUBJECT	AVERAGE SCORE		GAP	RIGHT	WRONG	SKIPPED
Maths out of 50 = 60 minus Q13 (4), Q14 (3), Q17 (3)		≈ 45.7 ≈ 37.2	8.5 per student	91% vs 76%	6% vs 23%	2% vs 1%
Physics out of 56 = 60 minus Q29 (4)		≈ 41.7 ≈ 25.8	15.9 per student	74% vs 50%	24% vs 44%	2% vs 6%
Chemistry out of 57 = 60 minus Q50 (3)		≈ 50.0 ≈ 36.7	13.3 per student	86% vs 68%	13% vs 28%	1% vs 4%

"Out of" = the subject's marks on this paper minus the questions left out by the final key (marks of each left-out question in brackets). Percentages are of all answers in that subject (questions × students), institute first, ours second.

Key takeaways

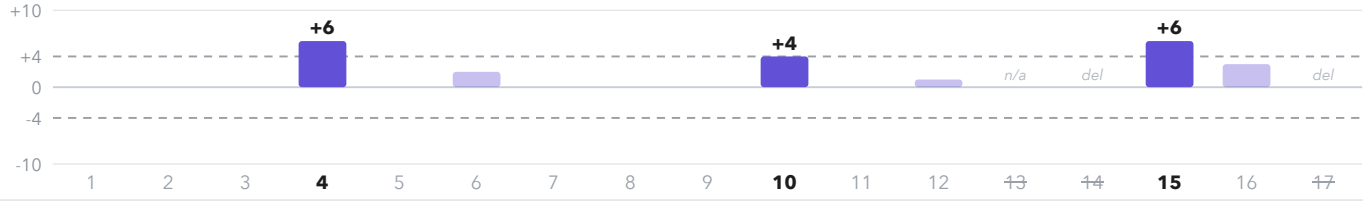
1. **Physics** is where we fall furthest behind: ≈ **15.9** marks per student (42% of the total gap).
2. **11 questions** were answered correctly by at least 4 more of the institute's top 10 than ours. Biggest: Q4 (10 vs 4), Q15 (9 vs 3), Q30 (10 vs 4).
3. We also got more wrong in **Physics**: 44% of answers wrong vs 24% – accuracy, not just coverage.
4. Where we did better: Q34 (5 vs 3).

2 Question-by-question gap

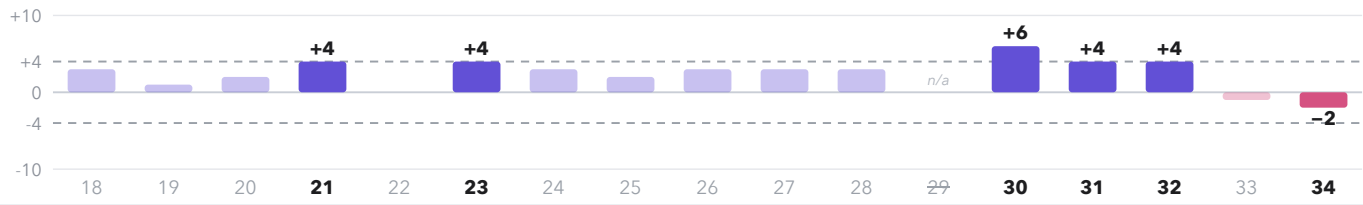
For each question: how many more of the institute's top 10 got it right than ours. Bars above the line = the institute did better; below = we did better. Gaps of 4+ (dashed line) are labelled.

Institute top 10 got more right · Our top 10 got more right · ---- Gap of 4 · "del" / "n/a" = left out (final key)

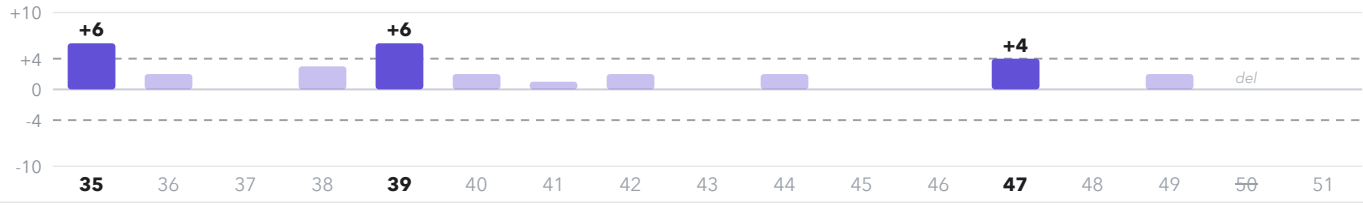
Maths — 3 questions with a gap of 4+ · gap ≈ 8.5 marks per student



Physics — 5 questions with a gap of 4+ · gap ≈ 15.9 marks per student



Chemistry — 3 questions with a gap of 4+ · gap ≈ 13.3 marks per student



3 Chapter gaps

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

CHAPTER	QS	INSTITUTE VS OURS	GAP PER STUDENT	GAP QUESTIONS
1. Physics gap ≈ 15.9 marks per student · 5 questions with a gap of 4+				
Laws of Motion	4	10.2 vs 4.0	≈ 6.2	Q23
Vectors	5	13.0 vs 7.5	≈ 5.5	Q31, Q32
Kinematics	7	18.5 vs 14.3	≈ 4.2	Q21, Q30
2. Chemistry gap ≈ 13.3 marks per student · 3 questions with a gap of 4+				
Mole Concept & Basics	9	29.6 vs 20.9	≈ 8.7	Q35, Q39, Q47
Atomic Structure	4	10.4 vs 7.6	≈ 2.8	–
Chemical Bonding	1	3.0 vs 1.8	≈ 1.2	–
Periodic Classification	2	7.0 vs 6.4	≈ 0.6	–
3. Maths gap ≈ 8.5 marks per student · 3 questions with a gap of 4+				
Logarithms	4	11.2 vs 7.9	≈ 3.3	Q4
Quadratic Equations	6	20.2 vs 17.2	≈ 3.0	Q10
Sequences & Series	4	14.3 vs 12.1	≈ 2.2	Q15

4 Biggest gaps – question list

Questions where at least 4 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5.

Maths

3 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q4	**Logarithms** – Logs in AP – solve for x	Single correct · +3 / −1	10/10	4/10 · 6 · 0	+6
Q15	**Sequences & Series** – AP: ratio of sums of 14 terms	Paragraph · +3 / 0	9/10	3/10 · 5 · 2	+6
Q10	**Quadratic Equations** – Roots of two linked quadratics	Integer · +4 / 0	8/10	4/10 · 6 · 0	+4

Physics

5 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q30	**Kinematics** – a from v(t) at a given position	Integer · +4 / 0	10/10	4/10 · 6 · 0	+6
Q21	**Kinematics** – Retardation from the t-x relation	Single correct · +3 / −1	10/10	6/10 · 3 · 1	+4
Q23	**Laws of Motion** – Block on block, no friction – motion of m	Multiple correct · +4 / −2	9/10	5/10 · 4 · 1	+4
Q31	**Vectors** – Carrom passage – displacement centre → edge	Paragraph · +3 / 0	4/10	0/10 · 10 · 0	+4
Q32	**Vectors** – Carrom passage – displacement edge → hole	Paragraph · +3 / 0	4/10	0/10 · 10 · 0	+4

Chemistry

3 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q35	**Mole Concept & Basics** – Limiting reagent over two-step reaction	Single correct · +3 / −1	10/10	4/10 · 3 · 3	+6
Q39	**Mole Concept & Basics** – Meaning of 1 gram-atom of nitrogen	Multiple correct · +4 / −2	10/10	4/10 · 6 · 0	+6
Q47	**Mole Concept & Basics** – Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)	Integer · +4 / 0	10/10	6/10 · 4 · 0	+4

Where our top 10 did better

Q34 5 vs 3 right

How to read this report. "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G8 (EXI) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.

Final key applied. Deleted in the final key: Q14, Q17, Q50. Not deleted in the final key, but the final marks sheet has no right / wrong / skipped entry for any student, so they cannot be counted: Q13 (key 4046), Q29 (key 9). Answer revised in the final key and counted as marked in the final marks sheet: Q25 (key 1 OR 2), Q48 (key 2 OR 5), Q51 (key 3). Left-out questions are marked "del" or "n/a" in the charts and excluded from both groups.

5 Maths – biggest gaps

3 questions, largest gap first
cropped from the question paper

Q4 — Logarithms – Logs in AP – solve for x

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **4 right**, 6 wrong, 0 skipped
Maths · Single correct · marking +3 / −1

Gap +6

4. If $\log_3 2, \log_3\left(2^x-5\right)$ & $\log_3\left(2^x-\frac{7}{2}\right)$ are in AP then the value of x is?

A)2 B)3 C)4 D)2&3

Ours wrong: Kolukula Hema Shiva Naga Hariharan, Karthikeya Silver, Sriyan Brahmakanti, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –

Q15 — Sequences & Series – AP: ratio of sums of 14 terms

Institute top 10: **9 right**, 1 wrong, 0 skipped · Our top 10: **3 right**, 5 wrong, 2 skipped
Maths · Paragraph · marking +3 / 0

Gap +6

Sum of n terms of two different Arithmetic progressions are in the ratio 5n+4:9n+6 then
15. The ratio of the sum of their first 14 terms is Q, then [100Q] is _____(where [.] is G.I.F)

Ours wrong: Ruperao Shritan, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu

Ours skipped: Nakka Mahidhar, Macharla Srivyshnav

Q10 — Quadratic Equations – Roots of two linked quadratics

Institute top 10: **8 right**, 1 wrong, 1 skipped · Our top 10: **4 right**, 6 wrong, 0 skipped
Maths · Integer · marking +4 / 0

Gap +4

10. Let a and b be roots of the equation $x^2-10cx-11d=0$ and the roots of $x^2-10ax-11b=0$ are c and d; then the value of $\frac{a+b+c+d}{605}$ is____?(Given $a\neq b\neq c\neq d\neq 0$)

Ours wrong: Kavuru Sahithi, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –

Physics – biggest gaps

5 questions, largest gap first
cropped from the question paper

Q30

Kinematics – a from v(t) at a given position

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **4 right**, 6 wrong, 0 skipped

Physics · Integer · marking +4 / 0

Gap +6

30. A body starts from origin and moves along x axis such that at any instant velocity is $v_t=4t^3-2t$ where t is in second and v_t is in ms^{-1}. The acceleration of the particle when it is 2m from the origin is $(20+K)\,ms^{-2}$, where K is

Ours wrong: Kolukula Hema Shiva Naga Hariharan, Fareeha Shaik, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar

Ours skipped: –

Q21

Kinematics – Retardation from the t-x relation

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **6 right**, 3 wrong, 1 skipped

Physics · Single correct · marking +3 / −1

Gap +4

21. What is the retardation of a moving particle if the relation between time (t) and position (x) is $t=Ax^2+Bx$, where A and B are appropriate constants.

A) $2A/(2Ax+B)^3$

B) $2B/(2Ax+B)^3$

C) $-2A/(2Ax+B)^3$

D) $-2B/(2Ax+B)^3$

Ours wrong: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu

Ours skipped: Macharla Srivyshnav

Q23

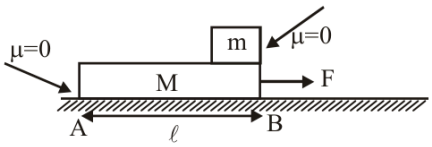
Laws of Motion – Block on block, no friction – motion of m

Institute top 10: **9 right**, 1 wrong, 0 skipped · Our top 10: **5 right**, 4 wrong, 1 skipped

Physics · Multiple correct · marking +4 / −2

Gap +4

23. In the figure small block is kept on M, then



A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

B) the acceleration of m w.r.t. ground is zero

C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

Ours wrong: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu, Nakka Mahidhar

Ours skipped: Macharla Srivyshnav

Q31

Vectors – Carrom passage – displacement centre → edge

Institute top 10: **4 right**, 5 wrong, 1 skipped · Our top 10: **0 right**, 10 wrong, 0 skipped

Physics · Paragraph · marking +3 / 0

Gap +4

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.

Answer the following question.

31. The magnitude of the displacement of the queen from the centre to the front edge is $\frac{2}{3}\sqrt{a}\,ft$, the value of a is _____

Ours wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –

Q32

Vectors – Carrom passage – displacement edge → hole

Institute top 10: **4 right**, 5 wrong, 1 skipped · Our top 10: **0 right**, 10 wrong, 0 skipped

Physics · Paragraph · marking +3 / 0

Gap +4

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.

Answer the following question.

32. The magnitude of the displacement of the queen from the front edge to the hole is $\frac{4}{3}\sqrt{b}\,ft$, the value of b is_____

Ours wrong: Ruperao Shritan, Kolukula Hema Shiva Naga Hariharan, Kavuru Sahithi, Fareeha Shaik, Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –

Chemistry – biggest gaps

3 questions, largest gap first
cropped from the question paper

Q35

Mole Concept & Basics – Limiting reagent over two-step reaction

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **4 right**, 3 wrong, 3 skipped

Chemistry · Single correct · marking +3 / −1

Gap +6

35. 5 moles of A, 6 moles of Z and mixed with sufficient amount of C to produce final product F. How many maximum moles of 'F' can be produced as per the given sequence of reaction ?

$A + 2Z \rightarrow B$; $B + C \rightarrow Z + F$

(A) 3 (B) 2 (C) 4 (D) 5

Ours wrong: Ruperao Shritan, Sriyan Brahmakanti, G Umesh Babu

Ours skipped: Karthikeya Silver, Nakka Mahidhar, Macharla Srivyshnav

Q39

Mole Concept & Basics – Meaning of 1 gram-atom of nitrogen

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **4 right**, 6 wrong, 0 skipped

Chemistry · Multiple correct · marking +4 / −2

Gap +6

39. 1 g atom of nitrogen represents:

(A) 6.02×10^{23} N_2 molecules (B) 22.4 litre of N_2 at N.T.P.

(C) 11.2 litre of N_2 at N.T.P. (D) 14 g of nitrogen

Ours wrong: Karthikeya Silver, Sriyan Brahmakanti, Chitluri Nayan Swaroop, G Umesh Babu, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –

Q47

Mole Concept & Basics – Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)

Institute top 10: **10 right**, 0 wrong, 0 skipped · Our top 10: **6 right**, 4 wrong, 0 skipped

Chemistry · Integer · marking +4 / 0

Gap +4

47. For $2H_2 + O_2 \rightarrow 2H_2O$, 8 g H_2 reacts with 32 g O_2. Find the mass of water formed.

Ours wrong: Karthikeya Silver, Chitluri Nayan Swaroop, Nakka Mahidhar, Macharla Srivyshnav

Ours skipped: –